TUGAS 5

No.

Date:

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Jawaban

1 Mencari nilai turunan

(a) $f(x) = \left(\frac{1}{x^2} - \frac{3}{x^4}\right)(x + 5x^3)$

$f'(x) = \left(\frac{d}{dx}\frac{1}{x^2} - \frac{3}{x^4}\right)(x+5x^3) + \left(\frac{d}{dx}x+5x^3\right)\left(\frac{1}{x^2} - \frac{3}{x^4}\right)$

$= \left(\frac{-2}{x^3} + \frac{12}{x^5}\right)(x+5x^3) + (1+15x^2)\left(\frac{1}{x^2} - \frac{3}{x^4}\right)$

$= \left(-\frac{2}{x^2} + \frac{12}{x^4} - 10 + \frac{60}{x^2}\right) + \left(\frac{1}{x^2} + 15 - \frac{3}{x^4} - \frac{45}{x^2}\right)$

$= \boxed{5 + \frac{14}{x^2} + \frac{9}{x^4}}$

(b) $f(t) = \frac{\sin t}{1 + \tan t}$

$f'(t) = \frac{\left(\frac{d}{dt}\sin t\right).(1+\tan t) - \sin t\left(\frac{d}{dt}(1+\tan t)\right)}{(1+\tan t)^2}$

$= \frac{(\cos t + \sin t) - \sin t . \sec^2 t}{1 + 2\tan t + \tan^2 t}$

$= \boxed{\frac{\cos t + \sin t - \sin t . \sec^2 t}{\sec^2 t + 2\tan t}}$

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(c) $f(x) = \dfrac{(4x+3)^4}{(x+1)^3}$

$f'(x) = \dfrac{\left(\frac{d}{dx}(4x+3)^4\right)\cdot(x+1)^3 - \left(\frac{d}{dx}(x+1)^3\right)\cdot(4x+3)^4}{\left((x+1)^3\right)^2}$

$= \dfrac{16(4x+3)^3(x+1)^3 - 3(x+1)^2(4x+3)^4}{(x+1)^6}$

$= \dfrac{16(4x^2+7x+3)^3 - 3\left((x+1)(4x+3)^2\right)^2}{(x+1)^6}$

$= \boxed{\dfrac{16(4x^2+7x+3)^3 - 3(16x^3+40x^2+33x+9)^2}{(x+1)^6}}$

atau (kanselasi $(x+1)$)

$= \boxed{\dfrac{(4x+3)^3(4x+7)}{(x+1)^4}}$

(d) $f(x) = \sqrt{\sin(1+x^2)}$

$f'(x) = \dfrac{1}{2} \cdot \dfrac{1}{\sqrt{\sin(1+x^2)}} \cdot \left(\frac{d}{dx}\sin(1+x^2)\right)$

$= \dfrac{1}{2\sqrt{\sin(1+x^2)}} \cos(1+x^2)\cdot 2x$

$= \boxed{\dfrac{x\cos(1+x^2)}{\sqrt{\sin(1+x^2)}}}$

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2. $f(x) = |x^2 - 4|$

(a) Untuk sebuah fungsi berbentuk $|g(x)|$, turunannya tidak ada pada $\forall x \ni g(x) = 0$ karena lancip.

Dengan demikian,

$g(x) = 0 \Rightarrow x^2 - 4 = 0$

$\Rightarrow (x+2)(x-2) = 0$

$\Rightarrow x = \{-2, 2\}$

Oleh karena itu, $|x^2-4|$ mempunyai turunan pada

$$\boxed{\begin{array}{l} X = \{x \mid x \in \mathbb{R}, x \neq 2, x \neq -2\} \\ \text{atau } \mathbb{R} \setminus \{-2, 2\} \end{array}}$$

Rumus turunannya dapat dipecah dengan

$$|x^2-4| = \begin{cases} x^2 - 4, & |x| \geq 2 \\ 4 - x^2, & |x| < 2 \end{cases}$$

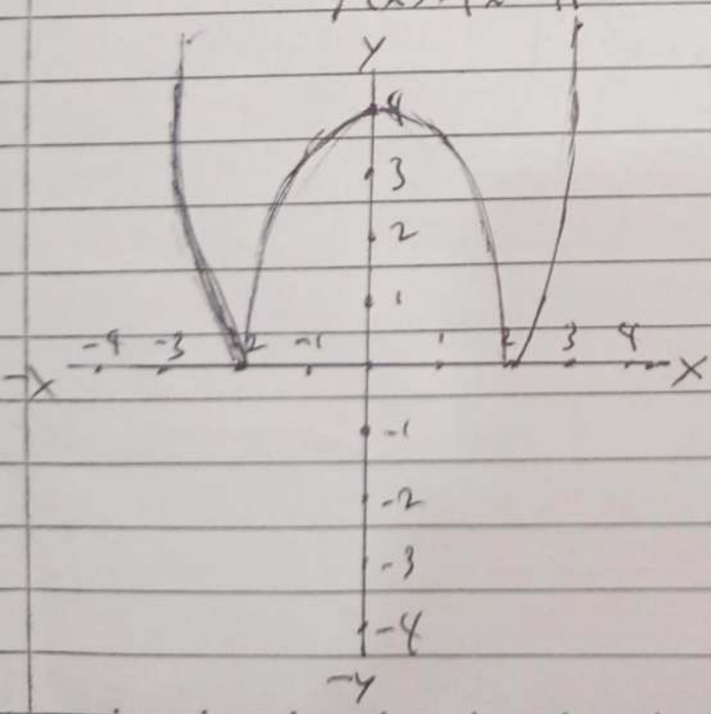
Maka turunannya

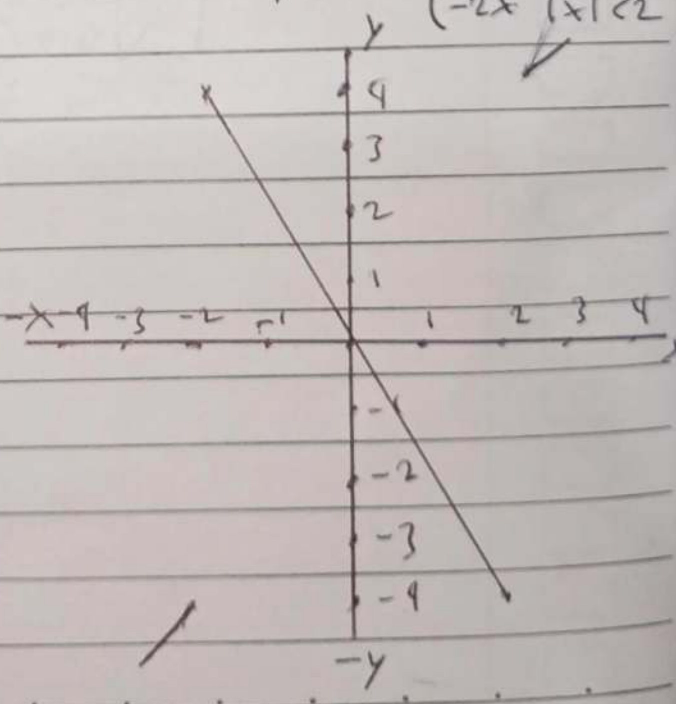
$$\boxed{\frac{d}{dx}|x^2-4| = \begin{cases} 2x, & x \in (-\infty, 2] \cup [2, \infty) \\ -2x & x \in (-2, 2) \end{cases}}$$

(b) Menggambar grafik $f(x)$ dan $f'(x)$

$f(x) = |x^2 - 4|$

$$f'(x) = \begin{cases} 2x, & |x| \geq 2 \\ -2x & |x| < 2 \end{cases}$$





Big BOSS

3. $f(1) = 4,\ g(1) = 1$

$f'(1) = 2,\ g'(1) = 6$

$h(x) = \dfrac{f(x)}{g(x) - f(x)}$

$h'(1)$?

$h'(x) = \dfrac{f'(x)(g(x) - f(x)) - f(x)(g'(x) - f'(x))}{(g(x) - f(x))^2}$

$\Rightarrow h'(1) = \dfrac{f'(1)(g(1) - f(1)) - f(1)(g'(1) - f'(1))}{(g(1) - f(1))^2}$

$\Rightarrow \quad = \dfrac{2(1-4) - 4(6-2)}{(1-4)^2}$

$= \dfrac{2(-3) - 4(4)}{3^2}$

$= \boxed{\dfrac{-22}{9}}$

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4. $y(x) = (x^2+1)(\sin(2x) - 1);\ (0,-1)$

$y'(x) = (2x)(\sin(2x) - 1) + (x^2+1)(2\cos(2x))$

$y'(0) = 2 \Rightarrow m = 2$ untuk $(0,-1)$

$\Rightarrow$ Garis singgungnya adalah

$\boxed{y = 2x - 1}$

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5. $f(x) = \begin{cases} x^2 & x \le 2 \\ mx + c & x > 2 \end{cases}$

Karena x^2 dan $mx+c$ kontinu, agar f memiliki turunan di seluruh bilangan real, turunan dari kiri dan dari kanan dari $f(x)$ harus sama, maka

$f'(x) = \begin{cases} 2x & x \le 2 \\ m & x > 2 \end{cases}$

Maka, agar $\lim_{x \to 2^-} f'(x) = \lim_{x \to 2^+} f(x)$, $m = 2(2)$

$= 4$.

Dengan demikian, bentuk fungsinya

$$f(x) = \begin{cases} x^2 & x \le 2 \\ 4x+c & x > 2 \end{cases}$$

Karena $f(x)$ punya turunan di $x=2$,

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x)$$

$\Rightarrow [x=2] \quad x^2 = 4x + c$

$\Rightarrow \quad 4 = 8 + c$

$\Rightarrow \quad c = -4$

Oleh karena itu

$\boxed{m = 4 \text{ dan } c = -4}$

6. $f(1) = \frac{\pi}{4} \cdot f'(1) = \frac{1}{2}$

$F(x) = \tan^2(f(x))$

$F'(x) = f'(x)\sec^2(f(x)) \cdot 2\tan(f(x))$

$\Rightarrow F'(1) = f'(1)\sec^2(f(1)) \cdot 2 \cdot \tan(f(1))$

$= \frac{1}{2} \cdot \sec^2\left(\frac{\pi}{4}\right) \cdot 2 \cdot \tan\left(\frac{\pi}{4}\right)$

$= \frac{1}{2} \cdot 2 \cdot 2 \cdot 1$

$= \boxed{2}$